

For a given function to compute n inputs, the divide and conquer strategy splits input into k distinct subsets when k is between $1 < k \leq n$, yielding the k subproblems. These subproblems must be solved and a method must be found to combine the solution of sub-problems into a solution of a given function.

* CONTROL ABSTRACTION OF DIVIDE AND CONQUER :-

It is nothing but the algorithm description of divide and conquer.

Algorithm DAND C (P) divide and conquer
 $\{$ $\rightarrow$ problem
 if small (P) then return S(P);
 else $\{$
 divide P into smaller instances $P_1, P_2, \dots, P_k$ $k \geq 1$
 Apply DAND C to each of these subproblems;
 return combine (DAND C, DAND C (P_1), ..., DAND C (P_k));
 }
 $\}$

The complexity of many divide & conquer algorithm is given by Recurrence of the form $T(n) = \begin{cases} T(1) & n=1 \\ aT(n/b) + f(n) & n>1 \end{cases}$ where $a \in b$ are constants and $n = b^k$,

$$k = \log_b n$$

In order to find the time complexities for any divide and conquer problems, we have two techniques

- i) Substitution Method
- ii) Recurrence / Master Method.

i) SUBSTITUTION METHOD

In substitution method, the following formulas and a tabular column is used to solve any kind of problem.

$$(1) h(n) = \frac{f(n)}{n^{\log_b a}}$$

$$(2) T(n) = n^{\log_b a} [T(1) + \mu(n)]$$

universal length value

$h(n)$	$\mu(n)$
$O(n^r), r < 0$	$O(1)$
$O((\log n)^i), i \geq 0$	$O((\log n)^{i+1}/i+1)$
$\Omega(n^r), r > 0$	$\Theta(h(n))$

$$1) a=1, b=2, f(n)=c$$

$$h(n) = \frac{c}{n^{\log_2 1}} = \frac{c}{n^0} = c = c(1)^0 = c$$

$$\mu(n) = \Theta((\log n)^{0+1}/0+1) \quad T(n) = n^{\log_2 1} [T(1) + \mu(n)]$$

$$\mu(n) = \Theta(\log n)$$

$$= n^{\log_2 1} [T(1) + \Theta(\log n)]$$

$$= n^0 [T(1) + \Theta(\log n)]$$

$$T(n) = T(1) + \Theta(\log n)$$

$\therefore$ according to asymptotic notations,

$$T(1) = \text{constant}$$

$$T(n) = \Theta(\log n)$$

$$2) a=5, b=4, f(n)=cn^2$$

$$h(n) = \frac{cn^2}{n^{\log_4 5}} = \frac{cn^2}{n^{1.16}} = c n^{0.84}$$

$$\mu(n) = \Theta(h(n))$$

$$= \Theta(n^{0.84})$$

$$T(n) = n^{\log_4 5} [T(1) + \mu(n)] = n^{\log_4 5} [T(1) + \Theta(n^{0.84})]$$

$$\begin{aligned}
 &= n^{1.16} [T(1) + \Theta(n^{0.84})] \\
 &= n^{1.16} [\Theta(n^{0.84})] + T(1)(n^{1.16}) \\
 &= \Theta(n^{1.16(0.84)}) = \Theta(n^2) + n^{1.16} \quad \boxed{T(n) = \Theta(n^2) + n^{1.16}}
 \end{aligned}$$

3) $a=7$ $b=2$ $f(n) = 18n^2$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{18n^2}{n^{\log_2 7}} = \frac{18n^2}{n^{2.8}} = 18n^{2-2.8} = 18n^{-0.8}$$

$$\begin{aligned}
 \mu(n) &= \Theta(1) \\
 T(n) &= n^{\log_b a} [T(1) + \mu(n)] \\
 &= n^{\log_2 7} [T(1) + \Theta(1)]
 \end{aligned}$$

($\because T(1)$ = constant according to asymptotic notations)

$$\boxed{T(n) = n^{2.8}}$$

4) $a=9$ $b=3$ $f(n) = 4xn^6$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{4xn^6}{n^{\log_3 9}} = \frac{4xn^6}{n^2} = 4n^4 = \mu(n) = \Theta(h(n)) = \Theta(n^4)$$

$$T(n) = n^{\log_b a} [T(1) + \mu(n)]$$

$$= n^2 [T(1) + \Theta(n^4)]$$

$$= n^2 [\Theta(n^4)]$$

($\because T(1)$ = constant according to the asymptotic notations)

$$\boxed{T(n) = \Theta(n^6)}$$

5) $a=28$ $b=3$ $f(n) = cn^3$

6) $a=2$, $b=2$, $T(1)=2$, $f(n)=n$

7) $a=2$, $b=2$ $f(n)=cn$

8) $a=28$ $b=3$ $f(n)=cn^3$

$$h(n) = \frac{cn^3}{n^{\log_3 28}} = \frac{cn^3}{n^3} = c \quad \mu(n) = \Theta(\log n)$$

$$T(n) = n^3 [T(1) + \theta(\log n)]$$

$$= \theta(n^3 \log n)$$

6) $a=2$ $b=2$ $T(1)=2$ $f(n)=n$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{n}{n^{\log_2 2}} = \frac{n}{n} = 1 = 1 ()^0 = 1$$

$$h(n) = \theta((\log n)^1)$$

$$T(n) = n^{\log_b a} [T(1) + \mu(n)]$$

($\because T(1) = \text{constant}$ according to asymptotic Notations)

$$= n^{\log_2 2} [T(1) + \theta(\log n)]$$

$$= n [T(1) + \theta(\log n)] = n\theta(\log n) + 2n$$

$$T(n) = \theta(n \log n)$$

7) $f(n)=cn$ $a=2$ $b=2$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{cn}{n^{\log_2 2}} = \frac{cn}{n} = c ()^0 = c$$

$$\mu(n) = \theta(\log n)$$

$$T(n) = n^{\log_2 2} [T(1) + \mu(n)]$$

($\because T(1) = \text{constant}$ according to asymptotic Notations)

$$= n [T(1) + \theta(\log n)] = \theta(n \log n)$$

5) $a=2$ $b=3$ $f(n)=cn^3$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{cn^3}{n^{\log_3 2}} = \frac{cn^3}{n^{0.631}} = cn^{2.369} > 0$$

$$\mu(n) = O(1)$$

$$T(n) = n^{\log_3 2} [T(1) + O(1)]$$

($\because (T(1), O(1))$ are constants according to asymptotic notations)

$$T(n) = n^{3.033}$$

ii) RECURSIVE METHOD

i) $a=2 \quad b=2 \quad f(n)=n$

$$T(n) = aT(n/b) + f(n)$$

$$T(n) = 2T(n/2) + n \quad - (1)$$

$$T(n/2) = 2T(n/4) + n/2$$

$$T(n/4) = 2T(n/8) + n/4 \quad - (2)$$

Substitute (2) in (1)

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$= 4T(n/4) + 2 \cdot n/2 + n$$

$$T(n) = 4T(n/4) + 2n \quad - (3)$$

$$T(n/4) = 2T(n/8) + n/4 \quad (\text{put } n=n/4 \text{ in equation (1) and calculate } T(n/4))$$

$$T(n/8) = 2T(n/16) + n/8 \quad - (4)$$

Substitute (4) in (3)

$$T(n) = 4[2T(n/8) + n/4] + 2n$$

$$= 8T(n/8) + 4 \cdot n/4 + 2n$$

$$T(n) = 8T(n/8) + 3n \quad - (5)$$

⋮

$$T(n) = 2^k T(n/2^k) + k \cdot n$$

We know that

$$n = b^k$$

$$k = \log_b n$$

$$n = 2^k \Rightarrow k = \log_2 n$$

$$T(n) = 2^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + n \cdot \log_2 n$$

$$= n \log_2 n T(1) + n \log_2 n$$

$$T(n) = n + n \log_2 n$$

$$\therefore T(n) = \Theta(n \log_2 n)$$

$$2) a=2 \quad b=2 \quad f(n)=cn$$

$$T(n) = aT(n/b) + f(n)$$

$$T(n) = 2T(n/2) + cn \quad \text{--- (1)}$$

$$T(n/2) = 2T(n/2 \cdot 1/2) + \frac{cn}{2}$$

$$T(n/2) = 2T(n/4) + \frac{cn}{2}$$

$$T(n) = 2 \left[2T(n/4) + \frac{cn}{2} \right] + cn$$

$$= 4T(n/4) + cn + cn \Rightarrow T(n) = 4T(n/4) + 2cn \quad \text{--- (2)}$$

$$T(n/4) = 2T(n/4 \cdot 1/2) + \frac{cn}{4}$$

$$T(n/4) = 2T(n/8) + \frac{cn}{4}$$

$$T(n) = 4 \left[2T(n/8) + \frac{cn}{4} \right] + 2cn$$

$$T(n) = 8T(n/8) + 3cn \quad \text{--- (3)}$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + k(cn)$$

$$n = b^k \quad k = \log_b n$$

$$n = 2^k \quad k = \log_2 n$$

$$T(n) = 2^{\log_2 n} T\left(\frac{2^k}{2^k}\right) + (cn)(\log_2 n)$$

$$T(n) = n^{\log_2 2} T(1) + (cn)(\log_2 n)$$

$$= nT(1) + cn \log_2 n$$

$$T(n) = n + cn \log_2 n$$

$$T(n) = \underline{cn \log_2 n} \Rightarrow$$

$$T(n) = \Theta(n \log_2 n)$$

3) $a=1$ $b=2$ $f(n) = cn$

$$T(n) = aT(n/b) + f(n)$$

$$T(n) = T(n/2) + cn$$

$$T(n/2) = T(n/4) + \frac{cn}{2}$$

$$T(n) = \left[T(n/4) + \frac{cn}{2} \right] + cn$$

$$T(n) = T(n/4) + \frac{3cn}{2} \quad \text{--- (1)}$$

$$T(n/4) = T(n/8) + \frac{cn}{4}$$

$$T(n/4) = T(n/8) + \frac{cn}{4}$$

$$T(n) = T(n/8) + \frac{cn}{4} + \frac{3cn}{2}$$

$$T(n) = T(n/8) + \frac{7cn}{4}$$

$$T(n) = T(n/8) + \frac{cn}{4} + \frac{3cn}{2} + \frac{7cn}{4}$$

$$= T(n/8) + c \left[\frac{n}{4} + \frac{n}{2} + \frac{n}{4} + \dots \right]$$

$$= T(n/8) + cn \left[\frac{1}{4} + \frac{1}{2} + \frac{1}{4} + \dots \right]$$

$$= T(n/8) + nc \left[\frac{1 - (1/2)^k}{1 - 1/2} \right]$$

$$= T(n/8) + 2nc \left[\frac{2^k - 1}{2^k} \right]$$

$$= T(n/2^k) + 2nc \left[\frac{2^k - 1}{2^k} \right]$$

$$= T\left(\frac{n}{2^k}\right) + 2 \cdot 2^k \cdot c \left[\frac{2^k - 1}{2^k} \right]$$

$$= T(1) + 2c[2^k - 1]$$

$$= T(1) + c[2^k - 1]$$

$$= T(1) + c \cdot 2^k - c$$

$$= c \cdot 2^k - c$$

Harar $a^2 + \dots$ and $r < 1$

$$= a(1 - r^k)$$

$$\text{if } r > 1 \quad \frac{1 - r^k}{1 - r}$$

$$\frac{a(r^k - 1)}{r - 1}$$

$$n = 2^k$$

$$n = 2^k$$

$$k = \log_2 n$$

$$n = 2^k$$

$$k = \log_2 n$$

$$T(n) = \Theta(n)$$

$$4) \quad a=2 \quad b=3 \quad f(n)=n^3$$

$$T(n) = aT(n/b) + f(n)$$

$$\boxed{T(n) = 2T(n/3) + n^3}$$

$$T(n/3) = 2T(n/3 \cdot 1/3) + (n/3)^3$$

$$T(n/3) = 2T(n/9) + (n/3)^3$$

$$T(n) = 2 \left[2T(n/9) + \frac{n^3}{3^3} \right] + n^3$$

$$T(n) = 4T(n/9) + 2(n/3)^3 + n^3 \quad \text{--- (1)}$$

$$T(n/9) = 2T(n/9 \cdot 1/3) + (n/9)^3$$

$$\textcircled{2} \text{ in } \textcircled{1} \quad = 2T(n/27) + \left(\frac{n^3}{3^2}\right)^3 \quad \text{--- (2)}$$

$$T(n) = 4 \left[2T(n/27) + \left(\frac{n}{3^2}\right)^3 \right] + 2\left(\frac{n}{3}\right)^3 + n^3$$

$$= 8T(n/27) + 2^2 \cdot \left(\frac{n}{3^2}\right)^3 + 2^1 \left(\frac{n}{3}\right)^3 + n^3$$

$$T(n) = 2^K T\left(\frac{n}{3^K}\right) + n^3 \left[2^2 \cdot \left(\frac{1}{3^2}\right)^3 + 2^1 \left(\frac{1}{3}\right)^3 + 2^0 \left(\frac{1}{3}\right)^3 + \dots \right]$$

$$= 2^K T\left(\frac{n}{3^K}\right) + n^3 \left[\text{constant} \right], \text{ can be neglected.}$$

$$= 2^K T\left(\frac{n}{3^K}\right) + n^3$$

$$n = b^K$$

$$n = 3^K$$

$$K = \log_3 n$$

$$T(n) = 2^K T\left(\frac{n}{3^K}\right) + n^3$$

$$T(n) = 2^{\log_3 n} T\left(\frac{3^{\log_3 n}}{3^{\log_3 n}}\right) + n^3$$

$$\boxed{T(n) = \Theta(n^3)}$$

($\therefore$ According to order of growth)
 $\log_3 n < n^3$

$$4) \quad a=4 \quad b=2 \quad f(n)=n^2$$

$$T(n) = aT(n/b) + f(n)$$

$$T(n) = 4T(n/2) + n^2$$

$$T(n/2) = 4T(n/4) + \frac{n^2}{2^2}$$

$$T(n/2) = 4T(n/4) + \frac{n^2}{2^2} \quad \text{--- (1)}$$

$$T(n/4) = 4T(n/8) + \frac{n^2}{4^2}$$

$$\textcircled{2} \text{ in } \textcircled{1} \quad T(n/4) = 4T(n/8) + \left(\frac{n}{4}\right)^2 \text{ --- (2)}$$

$$T(n) = 4^2 \left[4T(n/8) + \left(\frac{n}{4}\right)^2 \right] + 2n^2$$

$$= 4^3 T(n/8) + n^2 + 2n^2$$

$$= 4^3 T(n/8) + 3n^2$$

$$T(n) = 4^k T(n/2^k) + kn^2$$

$$n = 2^k \Rightarrow n = 2^k \Rightarrow k = \log_2 n$$

$$T(n) = 4^k T\left(\frac{2^k}{2^k}\right) + \log_2 n \cdot n^2$$

$$= 4^k T(1) + n^2 \log_2 n$$

$$T(n) = 4^k T(1) + n^2 \log_2 n = 4^{\log_2 n} T(1) + n^2 \log_2 n$$

$$T(n) = n^2 \log_2 n + 2n \quad (\because \text{According to order of growth } 2n < n^2 \log_2 n)$$

$$T(n) = \Theta(n^2 \log_2 n)$$

$$5. \quad T(n) = \begin{cases} k & n=1 \\ 3T(n/2) + kn & n>1 \end{cases}$$

$$T(n) = 3T(n/2) + kn$$

$$T(n/2) = 3T(n/4) + k \cdot \frac{n}{2}$$

$$T(n/2) = 3T(n/4) + k \cdot \frac{n}{2}$$

$$T(n/2) = 3T(n/4) + k \cdot \frac{n}{2}$$

$$T(n) = 3 \left[3T(n/4) + k \cdot \frac{n}{2} \right] + kn$$

$$= 9T(n/4) + \frac{3kn}{2} + kn$$

$$T(n/4) = 3T(n/8) + k \cdot \frac{n}{4} = 3T(n/8) + \frac{kn}{4}$$

$$T(n) = 9 \left[3T(n/8) + \frac{kn}{4} \right] + \frac{3kn}{2} + kn \quad (d^3) T(n) = (d^3) T(n)$$

$$= 3^3 T(n/2^3) + \frac{9kn}{4} + \frac{3kn}{2} + kn$$

$$T(n) = 3^k T\left(\frac{n}{2^k}\right) + kn \left[1 + \frac{3}{2} + \left(\frac{3}{2}\right)^2 + \dots + \left(\frac{3}{2}\right)^{k-1} \right] = (u/d) T$$

$$= 3^k T(n/2^k) + kn \left[\frac{\left(\frac{3}{2}\right)^k - 1}{\frac{3}{2} - 1} \right] = (u/d) T(n) = (u/d) T$$

$$= 3^k T(n/2^k) + 2kn \left[\left(\frac{3}{2}\right)^k - 1 \right]$$

$$= 3^k T\left(\frac{2^k}{2^k}\right) + 2kn \left(\frac{3}{2}\right)^k - 2kn$$

$$= 3^k T(1) + 2kn \left(\frac{3}{2}\right)^k - 2kn$$

$$= k \cdot 3^k + 2kn \left(\frac{3}{2}\right)^k - 2kn$$

$$= k \cdot 3^{\log_2 n} + 2kn \left(\frac{3}{2}\right)^{\log_2 n} - 2kn = (n) T$$

$$= k \cdot 3^{\log_2 n} + \frac{2kn}{2} \cdot \frac{\log_2 n}{\log_2 2} - 2kn$$

$$= k \cdot 3^{\log_2 n} + \frac{2kn}{2} \cdot \frac{\log_2 n}{\log_2 2} - 2kn$$

$$= 3^{\log_2 n} \left(k + \frac{2kn}{2} \right) - 2kn = (n) T$$

$$= 3^{\log_2 n} \cdot \frac{\log_2 n}{2} - 2kn$$

$$1 = n \quad \log_2 n = (n) T$$

6

$$T(n) = mT(n/2) + an^2 \quad \text{P.T. } T(n) = O(n^{\log m})$$

$$T(n) = mT(n/2) + an^2$$

$$T(n/2) = mT(n/2 \cdot 1/2) + a(n/2)^2$$

$$T(n/2) = mT(n/4) + a(n/2)^2$$

$$T(n) = m \left[mT(n/4) + a(n/2)^2 \right] + an^2$$

$$T(n) = m^2 T(n/4) + m a (n/2)^2 + a n^2 + \dots \quad (1)$$

$$T(n/4) = m T(n/4 \cdot 1/2) + a (n/4)^2$$

$$T(n/4) = m T(n/8) + a (n/4)^2 \quad (2)$$

$$T(n) = m^2 [m T(n/8) + a (n/4)^2] + m a (n/2)^2 + a n^2$$

$$= m^3 T(n/8) + m^2 a (n/4)^2 + m a (n/2)^2 + a n^2$$

$$= m^k T(n/2^k) + a \left[\left(\frac{n}{2}\right)^2 + m \left(\frac{n}{2}\right)^2 + m^2 \left(\frac{n}{2}\right)^2 + \dots \right]$$

$$= m^k T(n/2^k) + a n^2 \left[1 + \left(\frac{m}{2}\right) + \left(\frac{m}{2}\right)^2 + \dots \right]$$

$$= m^k T\left(\frac{n}{2^k}\right) + a n^2 \left(\frac{1 - (m/2)^k}{1 - m/2} \right)$$

$\frac{5}{4}$
 $\frac{1}{4}$

→ BINARY SEARCH

Divide and conquer solves the binary search problem in the following way:

If p is a problem which contains more than one element, it can be divided into new sub problems as follows:

→ Pick the index q in the range i, l and compare x with $a[q]$. there are 3 possibilities

$\Rightarrow x = a[q]$ the p is immediately solved.

= If $x < a[q]$ x has to be searched for only sublist a_i can $(a_i, a_{i+1}, \dots, a_{q-1})$ and p reduces to $(a_{q+1}, a_{q+2}, \dots, a_n)$ and

= If $x > a[q]$ the subset to be searched $a_{q+1}, \dots, a_n$ and p reduces to $p(l, q, a_{q+1}, \dots, a_n, x)$

The time taken for the above operations is $O(1)$

The Q is always chosen such that a_q is middle element i.e., $q = \frac{n+1}{2}$, the resultant search algorithm is called binary search.

ALGORITHM FOR BINARY SEARCH USING RECURSIVE AND NON RECURSIVE

i) Recursive Binary Search

Algorithm BinSrch(a, l, h, x)

```
{
    if (l == h) then { // If p is small
        if (x == a[l]) then return l;
        else return 0;
    }
    else {
        mid := (l+h)/2; // reduce p into sub problems
        if (x == a[mid]) then return mid;
        else if (x < a[mid]) then
            return BinSrch(a, l, mid-1, x);
        else return BinSrch(a, mid+1, h, x);
    }
}
```

ii) Iterative / Non Recursive Binary Search

Algorithm BinSearch(a, n, x)

```
{
    low := 1; high := n;
    while (low <= high) do {
        mid := (low+high)/2;
        if (x < a[mid]) then high := mid-1;
        else if (x > a[mid]) then
            low := mid+1;
        else return mid;
    }
}
```

1) $-15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151$

$x = 151$

low

high

mid

$x = 151$

$151 > 54$

1

14

$\frac{15}{2} = 7$

$\text{mid}[7] = 54$

$\text{low} := \text{mid} + 1$

8

14

$\frac{22}{2} = 11$

$x = 151$
 $\text{mid}[11] = 125$

$151 > 125$
 $\text{low} := \text{mid} + 1$

12

14

$\frac{26}{2} = 13$

$x = 151$
 $\text{mid}[13] = 142$

$151 > 142$
 $\text{low} := \text{mid} + 1$

14

14

$\frac{28}{2} = 14$

$x = 151$
 $\text{mid}[14] = 151$

$151 = 151$

element found.

$x = 9$

$x = -14$

low

high

mid

$x = 9$

$\text{high} := \text{mid} - 1;$

1

14

$\frac{15}{2} = 7$

$\text{mid}[7] = 54$

1

6

$\frac{7+6}{2} = 3$

$9 < 54$
 $x = 9$
 $\text{mid}[3] = 0$

$\text{low} := \text{mid} + 1;$

$9 > 0$

4

6

5

$x = 9$
 $\text{mid}[5] = 9$

$9 = 9$

↳ element found.

$x = -14$

low

high

mid

$x = -14$

$\text{high} := \text{mid} - 1;$

1

14

$\frac{15}{2} = 7$

$\text{mid}[7] = 54$

$-14 < 54$

1

6

3

$x = -14$
 $\text{mid}[3] = 0$

$\text{high} := \text{mid} - 1;$

$-14 < 0$

1

2

1

$x = -14$
 $\text{mid}[1] = -15$

$\text{high} := \text{mid} - 1;$
 $\text{low} := \text{mid} + 1;$

$-15 < -14$

2

2

2

$x = -14$
 $\text{mid}[2] = -6$

$\text{high} := \text{mid} - 1;$

$-14 < -6$

$$a = -14$$
$$\text{mid}[1] = -15$$
$$-14 > -15$$

mid[1] = -15

$$-14 > -15$$
$$\text{low} = \text{mid} + 1$$
$$a = -14$$
$$\text{mid}[1] = -18$$

low = mid + 1

-14 > -15

* Note :
→ If n is in the range $(2^{k-1}, 2^k)$ then binary search makes at most k element comparisons for a successful search

→ $(K-1, K)$ comparison for an unsuccessful search.

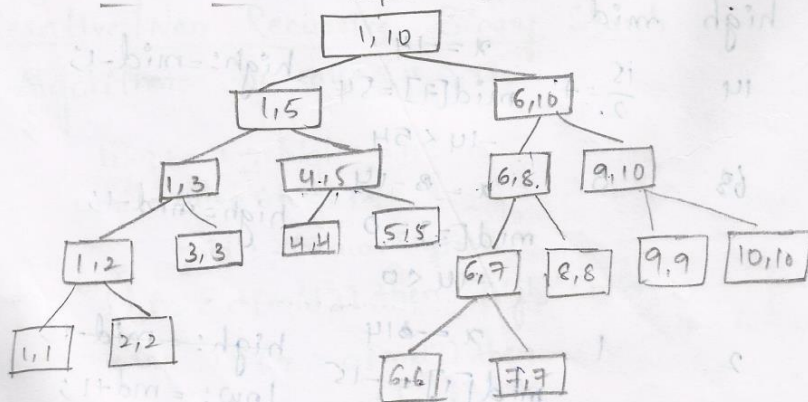
→ Time taken for ~~un~~successful search is $O(\log n)$

→ Time taken for unsuccessful search is $\Theta(\log n)$

→ MERGE SORT :

For a given sequence of n elements $a[1], \dots, a[n]$, the general idea is to split it into 2 sets i.e., $a[1]$ to $a[\frac{n}{2}]$ and $a[\frac{n}{2}+1] - \dots - a[n]$. Each set is individually sorted and the resulting sorted sequences are merged to produce a single sorted sequence of n elements. used in divide & con strategy

TREE CALLS OF MERGE SORT :



1) 310, 285, 179, 652, 351, 423, 861, 254, 450, 520

$a[1]$ $a[2]$ $a[3]$

The given array size elements are 1 to 10. So divide the n size by using $\text{mid} = \text{low} + \text{high} / 2$; i.e., $l=1$ and $h=10$ and $m=5$. $a[1] \dots a[5]$ is one set. $a[6]$ to $a[10]$ is another set. First sort the $a[1]$ to $a[5]$.

310 285 179 652 351
 $a[1]$ $a[2]$ $a[3]$ $a[4]$ $a[5]$

Compare $a[1]$ with $a[2]$

$a[1] > a[2]$. So merge $a[1]$ & $a[2]$.

285 310 $a[1,2]$

$a[1,2]$ is compared with $a[3]$
 So $a[1,2]$ is merged with $a[3]$ because
 $a[1,2] > a[3]$

179 285 310 $a[1,3]$

$a[1,3]$ is compared with $a[4,5]$

652 351

$a[4]$ $a[5]$

They are merged ($a[4] \times a[5]$)

a 351 652
 $a[4,5]$

179 285 310 351 652

→ Set ① with
 Sorted List

423 861 254 450 520
 $a[6]$ $a[7]$ $a[8]$ $a[9]$ $a[10]$

Compare $a[6]$ with $a[7]$. $a[6] < a[7]$

So no need to sort the elements

$a[6,7] = 423 \quad 861$

Now, $a[6,7]$ is compared with $a[8]$
 $a[6,7] > a[8]$. So merge, the list

254 423 861
 $a[6,8]$

$a[9]$ is merged with $a[10]$.

$a[9] < a[10]$. No merge.

$a[9,10] = 450 \quad 520$

Now $a[6,8]$ is merged with $a[9,10]$

254 423 861 | 450 520

254 423 450 520 861

→ Set ② with

Sorted list

Combine Set ① & Set ②

179, 254, 285, 310

520, 652

2. 33, 29, 17, 66, 37, 44, 78, 26, 51, 56

$a[1] \ a[2] \ a[3] \ a[4] \ a[5] \ a[6] \ a[7] \ a[8] \ a[9] \ a[10]$

$$\text{mid} = \frac{\text{low} + \text{high}}{2} = \frac{1 + 10}{2} = \frac{11}{2} = 5$$

33 29 17 66 37 | 44 78 26 51 56
 $a[1] \ a[2] \ a[3] \ a[4] \ a[5] \ a[6] \ a[7] \ a[8] \ a[9] \ a[10]$

$a[1] > a[2]$ So merge $a[1]$ and $a[2]$

$\underbrace{29 \ 33} \rightarrow a[1,2]$ is merged with $a[3]$

$\underbrace{17 \ 29 \ 33} \ a[1,3]$
 $a[4] > a[5]$ So merge $a[4]$ & $a[5]$

$\underbrace{37 \ 66} \rightarrow a[4,5]$
 $a[1,3]$ is merged with $a[4,5]$ to get $a[1,5]$

17 29 33 37 66

↳ set ① with the sorted list.

set ① & set ② are combined to give the required sorted elements.

17, 26, 29, 33, 37, 44, 51, 56, 66, 78 → are the required sorted elements.

The recurrence relation for merge sort is given as

$$T(n) = \begin{cases} a & \text{when } n=1 \\ 2T(n/2) + cn & n>1 \end{cases} \quad c \text{ is constant.}$$

$a[6] < a[7]$. No need of Merging.

$a[6,7] = 44 \ 78$

$a[6,7]$ is compared with $a[8]$

$\underbrace{26 \ 44 \ 78} \ a[6,8]$

$a[9] < a[10]$. So no need to merge

$a[9,10] = 51 \ 56$

$a[6,8]$ is merged with $a[9,10]$ to get $a[6,10]$

26 44 51 56 78

↳ set ② with the sorted list.

$$a=2 \quad b=2 \quad f(n)=cn$$

$$h(n) = \frac{f(n)}{n^{\log_b a}} = \frac{cn}{n^{\log_2 2}} = \frac{cn}{n} = c = c \cdot 1 = c$$

$$\mu(n) = \theta(\log n)^{0+1} = \theta(\log n)$$

$$T(n) = n^{\log_b a} [T(1) + \mu(n)]$$

$$= n^{\log_2 2} [a + \theta(\log n)]$$

$$= n [\theta(\log n)] = \theta(n \log n)$$

$$T(n) = \theta(n \log n)$$

The time complexity of a Merge Sort in best case, worst and average analysis is $O(n \log n)$

The recurrence relation for binary search is given as

$$T(n) = \begin{cases} a, & n=1 \\ T(n/2) + 1, & n>1 \end{cases}$$

Algorithm for Merge Sort :-

Algorithm Mergesort(low, high)

{ if (low < high) then

{

mid := low + high / 2;

MergeSort(low, mid);

MergeSort(mid+1, high);

MergeSort(low, mid+1, high);

}

Algorithm Mergesort(low, mid, high)

{

h := low, i := low, j := mid+1;

while (h ≤ mid) and (j ≤ high) do

{

if (a[h] ≤ a[j]) then

{ b[i] := a[h];

h = h+1;

}

else {

```

    b[i] = a[j];
    j := j + 1;
  }
  i := i + 1;
}
if (h > mid) then
  for (k := j to high) do
    {
      b[i] := a[k];
      i := i + 1;
    }
  else {
    for k := h to mid do
      b[i] := a[k];
      i := i + 1;
    }
    for (k := low to high) do
      a[k] := b[k];
  }
}

```

→ STRASSEN'S MATRIX MULTIPLICATION :

Let A, B be two matrix of $n \times n$, the product matrix $C = A \times B$ is also an $n \times n$ matrix whose i th & j th element is found by taking elements in i th row of a and j th row of b by multiplying them to get $c[i, j] = \sum A(i, k) \times B(k, j) \quad i \leq k \leq n$

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$C = A \times B = \begin{bmatrix} \underbrace{A_{11}B_{11} + A_{12}B_{21}}_{c_{11}} & \underbrace{A_{11}B_{12} + A_{12}B_{22}}_{c_{12}} \\ \underbrace{A_{21}B_{11} + A_{22}B_{21}}_{c_{21}} & \underbrace{A_{21}B_{12} + A_{22}B_{22}}_{c_{22}} \end{bmatrix}$$

$$T(n) = 8T(n/2) + an^2 \quad n > 2$$

The computing time of above matrix is $T(n) = 8T(n/2) + an^2$, $n > 2$, i.e., $T(n) = \Theta(n^3)$. The total computations it requires is 8 multiplications and 4 additions. In order to reduce the above computing time using divide & conquer policy Volker Strassen has introduced a method where you require 7 multiplications and 18 add/subtraction, i.e.,

$T(n) = 7T(n/2) + an^2$ H.W, $n > 2$, $T(n) = \Theta(n^{2.81})$. This method involves

computing of 7 $n/2 \times n/2$ matrices i.e.,

$$P = (A_{11} + A_{12})(B_{11} + B_{22})$$

$$Q = (A_{21} + A_{22})B_{11}$$

$$R = A_{11}(B_{12} - B_{22})$$

$$S = A_{22}(B_{21} - B_{11})$$

$$T = (A_{11} + A_{12})B_{22}$$

$$U = (A_{21} - A_{11})(B_{11} + B_{12})$$

$$V = (A_{12} - A_{22})(B_{21} + B_{22})$$

~~W~~

$$C_{11} = P + S - T + V$$

$$C_{12} = R + T$$

$$C_{21} = Q + S$$

$$C_{22} = P + R - Q + U$$

1) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$; $A_{11}=1, A_{12}=1, A_{21}=1, A_{22}=1$
 $B_{11}=0, B_{12}=0, B_{21}=1, B_{22}=1$

$$P = (1+1)(0+1)$$

$$P = 2+0 = 2$$

$$Q = (2)(0) = 0$$

$$R = 1(-1) = -1$$

$$S = 1(1) = 1$$

$$T = (2)(1) = 2$$

$$U = (1-1)(0+0) = 0$$

$$V = (1-1)(1+1) = 0$$

$$C_{11} = 2 + 1 - 2 + 0$$

$$C_{11} = +1$$

$$C_{12} = -1 + 2 = 1$$

$$C_{21} = 0 + 1 = 1$$

$$C_{22} = 2 - 1 - 0 + 0$$

$$= +1$$

$$C = \begin{bmatrix} +1 & 1 \\ 1 & +1 \end{bmatrix}$$

2)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 6 & 0 & 3 \\ 4 & 1 & 1 & 2 \\ 0 & 3 & 5 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 & 2 & 7 \\ 3 & 1 & 3 & 5 \\ 2 & 0 & 1 & 3 \\ 1 & 4 & 5 & 1 \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix}$$

$$A_{12} = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix}$$

$$B_{11} = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$$

$$B_{12} = \begin{bmatrix} 2 & 7 \\ 3 & 5 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix}$$

$$A_{22} = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix}$$

$$B_{21} = \begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix}$$

$$B_{22} = \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix}$$

$$P = \left(\begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} \right) \left(\begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix} \right)$$

$$P = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 7 \\ 8 & 2 \end{bmatrix} = \begin{bmatrix} 16 & 22 \\ 58 & 47 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \left(\begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 15 \\ 20 & 12 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \left(\begin{bmatrix} 2 & 7 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 10 \\ -12 & 24 \end{bmatrix}$$

$$P = (2+6)(4) = 8(4) = 32$$

$$Q = 2(20) = 40$$

$$R = 2(5) = 10$$

$$S = 6(6) = 36$$

$$T = 6(2) = 12$$

$$U = 3(9) = 27$$

$$V = (-2)(10) = -20$$

$$C_{11} = P + S - T + V$$

$$= 32 + 36 - 12 - 20$$

$$= 68 - 32$$

$$C_{11} = 36$$

$$C_{12} = R + T = 22$$

$$C_{21} = Q + S = 22 + 36 = 58$$

$$C_{22} = P + R - Q + U = 32 + 10 - 22 + 27$$

$$C_{22} = 69 - 22 = 47$$

$$P = \begin{bmatrix} 36 & 22 \\ 58 & 47 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \left[\begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 5 & 0 \end{pmatrix} \right] = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix}$$

$$P = (A_{11} + A_{22})(B_{11} + B_{22}) = (2)(8) = 16$$

$$Q = B_{11}(A_{21} + A_{22}) = 5(3+1) = 8$$

$$R = A_{11}(B_{12} - B_{22}) = 1(3-3) = 0$$

$$S = A_{22}(B_{21} - B_{11}) = 1(5-3) = 2$$

$$T = (A_{11} + A_{12})B_{22} = 3(5) = 15$$

$$U = (A_{21} - A_{11})(B_{11} + B_{12}) = (8)(2) = 16$$

$$V = (A_{12} - A_{22})(B_{21} + B_{22}) = 3(8) = 24$$

$$C_{11} = P + S - T + V = 16 + 2 - 15 + 24 = 27$$

$$C_{12} = R + T = 0 + 15 = 15$$

$$C_{21} = Q + S = 8 + 2 = 10$$

$$C_{22} = P + R - Q + U = 16 + 0 - 8 + 16 = 24$$

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \left[\begin{pmatrix} 2 & 7 \\ 3 & 5 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 5 & 1 \end{pmatrix} \right] = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ -2 & 4 \end{bmatrix}$$

$$P = (7)(5) = 35$$

$$Q = (1)(6) = 6$$

$$R = 1(0) = 0$$

$$S = 6(-3) = -18$$

$$T = (3)(4) = 12$$

$$U = (-1)(4) = -4$$

$$V = (-4)(2) = -8$$

$$2) A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 6 & 0 & 3 \\ 4 & 1 & 1 & 2 \\ 0 & 3 & 5 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 4 & 2 & 7 \\ 3 & 1 & 3 & 5 \\ 2 & 0 & 1 & 3 \\ 1 & 4 & 5 & 1 \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \quad A_{21} = \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix} \quad B_{11} = \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \quad B_{12} = \begin{bmatrix} 2 & 7 \\ 3 & 5 \end{bmatrix}$$

$$A_{12} = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix} \quad A_{22} = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} \quad B_{21} = \begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix} \quad B_{22} = \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix}$$

$$P = (A_{11} + A_{22})(B_{11} + B_{12})$$

$$= \left[\begin{pmatrix} 1 & 2 \\ 0 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 5 & 0 \end{pmatrix} \right] \left[\begin{pmatrix} 1 & 4 \\ 3 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 5 & 1 \end{pmatrix} \right]$$

$$P = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 & 7 \\ 8 & 2 \end{bmatrix} \quad a \times b$$

$$P = 4(2+6) = 4(8) = 32$$

$$Q = 2(11) = 22$$

$$R = 2(5) = 10$$

$$S = 6(6) = 36$$

$$T = 6(2) = 12$$

$$U = 3(9) = 27$$

$$V = (-2)(10) = -20$$

$$P = \begin{bmatrix} 36 & 22 \\ 58 & 47 \end{bmatrix}$$

$$Q = (A_{21} + A_{22})B_{11}$$

$$Q = \left[\begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 5 & 0 \end{pmatrix} \right] \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix}$$

$$P = (8)2 = 16$$

$$Q = 1(8) = 8$$

$$R = 5(3) = 15$$

$$S = 3(2) = 6$$

$$T = 1(8) = 8$$

$$U = 0$$

$$V = 0$$

$$Q = \begin{bmatrix} 14 & 23 \\ 14 & 23 \end{bmatrix}$$

$$C_{11} = P + S - T + V$$

$$C_{11} = 32 + 36 - 12 - 20 = 36$$

$$C_{12} = R + T$$

$$C_{12} = 10 + 12 = 22$$

$$C_{21} = Q + S$$

$$C_{21} = 22 + 36 = 58$$

$$C_{22} = P + R - Q + U = 32 + 10 - 22 + 27 = 47$$

$$Q_{11} = P + S - T + V = 16 + 6 - 8 + 0 = 14$$

$$C_{12} = R + T = 15 + 8 = 23$$

$$C_{21} = Q + S = 8 + 6 = 14$$

$$C_{22} = P + R - Q + U = 16 + 15 - 8 + 0 = 23$$

$$R = A_{11}(B_{12} - B_{22})$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \left(\begin{bmatrix} 2 & 7 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 12 \\ -12 & 24 \end{bmatrix}$$

$$p = 7(5) = 35$$

$$t = 3(4) = 12$$

$$C_{11} = 35 - 18 - 12 + 8 = -3$$

$$q = 6(1) = 6$$

$$u = (-1)5 = -5$$

$$C_{12} = 0 + 12 = 12$$

$$r = 1(0) = 0$$

$$v = (-4)2 = -8$$

$$C_{21} = 6 - 18 = -12$$

$$s = 6(-3) = -18$$

$$C_{22} = 35 + 0 - 6 - 5 = 24$$

$$R = \begin{bmatrix} -3 & 12 \\ -12 & 24 \end{bmatrix}$$

$$S = A_{22}(B_{21} - B_{11})$$

$$\begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} \left(\begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 4 \\ 3 & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} 1 & -4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5 & -20 \end{bmatrix}$$

$$p = (1+0)(1+3) = 4$$

$$C_{11} = p + s - t + v = 4 + 0 - 9 + 2 = -3$$

$$q = 5(1) = 5$$

$$C_{12} = R + T = -7 + 9 = 2$$

$$r = 1(-4-3) = -7$$

$$C_{21} = Q + S = 0 + 5 = 5$$

$$s = 0$$

$$C_{22} = R + T - P + R - Q + U$$

$$t = (1+2)3 = 9$$

$$= 4 - 7 - 5 - 12 = -20$$

$$u = 4(-3) = -12$$

$$v = 2(1) = 2$$

$$S = \begin{bmatrix} -3 & 2 \\ 5 & -20 \end{bmatrix}$$

$$T = (A_{11} + A_{12})B_{22}$$

$$\left(\begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 0 & 3 \end{bmatrix} \right) \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 31 & 18 \\ 45 & 9 \end{bmatrix}$$

$$p = (5+9)(1+1) = 2(14) = 28$$

$$C_{11} = p + s - t + v$$

$$q = (9)(1) = 9$$

$$= 28 + 36 - 8 - 36$$

$$r = 5(3-1) = 10$$

$$C_{11} = 20$$

$$s = 9(2) = 36$$

$$C_{12} = R + T = 10 + 8 = 18$$

$$t = (5+3)(1) = 8$$

$$C_{21} = Q + s = 9 + 36 = 45$$

$$u = 4(-5) = -20$$

$$C_{22} = p + R - Q + U = 28 + 10 - 9 - 20$$

$$v = 6(-6) = -36$$

$$= 88 - 20 - 9 = 18 - 9$$

$$C_{22} = 9$$

$$T = \begin{bmatrix} 31 & 18 \\ 45 & 9 \end{bmatrix}$$

$$U = (a_{21} - a_{11})(b_{11} + b_{12})$$

$$= \left[\begin{pmatrix} 4 & 1 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 0 & 6 \end{pmatrix} \right] \left[\begin{pmatrix} 1 & 4 \\ 3 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 7 \\ 3 & 5 \end{pmatrix} \right] = \begin{pmatrix} 3 & -1 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 3 & 11 \\ 6 & 6 \end{pmatrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} & b_{11} & b_{12} \\ a_{21} & a_{22} & b_{21} & b_{22} \end{bmatrix}$$

$$p = (3-3)(9) = 0$$

$$q = 3(0-3) = -9$$

$$r = 3(11-6) = 15$$

$$s = (-3)(3) = -9$$

$$t = (3-1)6 = 12$$

$$u = (0-3)(14) = -3(14) = -42$$

$$v = (-1+3)(12) = 2(12) = 24$$

$$C_{11} = P+S-T+V = 24-21=3$$

$$C_{12} = R+T = 15+12=28$$

$$C_{21} = Q+S = -9-9 = -18$$

$$C_{22} = P+R-Q+U = 0+15+9-42 = -18$$

$$U = \begin{bmatrix} 3 & 28 \\ -18 & -18 \end{bmatrix}$$

$$V = (a_{12} - a_{22})(b_{21} + b_{22})$$

$$V = \left[\begin{pmatrix} 3 & 4 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 5 & 0 \end{pmatrix} \right] \left[\begin{pmatrix} 2 & 0 \\ 1 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 5 & 1 \end{pmatrix} \right]$$

$$V = \begin{bmatrix} a_{11} & a_{12} & b_{11} & b_{12} \\ a_{21} & a_{22} & b_{21} & b_{22} \end{bmatrix}$$

$$p = 5(8) = 40$$

$$q = (-5+3)(3) = 3(-2) = -6$$

$$r = 2(3-5) = 2(-2) = -4$$

$$s = 3(3) = 9$$

$$t = 4(5) = 20$$

$$u = 6(-7) = -42$$

$$v = (-1)(11) = -11$$

$$C_{11} = P+S-T+V = 18$$

$$C_{12} = R+T = -4+20=16$$

$$C_{21} = Q+S = 3$$

$$C_{22} = P+R-Q+U = 40-4+6-42 = 0$$

$$= 40-4+6-42 = 0$$

$$= 40-4+6-42 = 0$$

$$V = \begin{bmatrix} 18 & 16 \\ 3 & 0 \end{bmatrix}$$

$$C_{11} = P+S-T+V = \begin{bmatrix} 36 & 22 \\ 58 & 47 \end{bmatrix} + \begin{bmatrix} -3 & 2 \\ 5 & -20 \end{bmatrix} - \begin{bmatrix} 3 & 18 \\ 45 & 9 \end{bmatrix} + \begin{bmatrix} 18 & 16 \\ 3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 17 & 22 \\ 21 & 18 \end{bmatrix}$$

$$\begin{array}{r} 36 \\ -3 \\ \hline 33 \\ -20 \\ \hline 13 \\ 18 \end{array}$$

$$C_{12} = R + T$$

$$= \begin{bmatrix} -3 & 12 \\ -12 & 24 \end{bmatrix} + \begin{bmatrix} 34 & 18 \\ 45 & 9 \end{bmatrix} = \begin{bmatrix} 31 & 30 \\ 33 & 33 \end{bmatrix}$$

$$C_{21} = Q + S$$

$$= \begin{bmatrix} 14 & 23 \\ 14 & 23 \end{bmatrix} + \begin{bmatrix} -3 & 2 \\ 5 & -20 \end{bmatrix} = \begin{bmatrix} 11 & 25 \\ 19 & 3 \end{bmatrix}$$

$$C_{22} = P + R - Q + U$$

$$= \begin{bmatrix} 36 & 22 \\ 58 & 47 \end{bmatrix} + \begin{bmatrix} -3 & 12 \\ -12 & 24 \end{bmatrix} - \begin{bmatrix} 14 & 23 \\ 14 & 23 \end{bmatrix} + \begin{bmatrix} 3 & 27 \\ -18 & -18 \end{bmatrix} = \begin{bmatrix} 22 & 39 \\ 14 & 30 \end{bmatrix}$$

$$C = \begin{pmatrix} \begin{bmatrix} 17 & 22 \\ 21 & 18 \end{bmatrix} & \begin{bmatrix} 31 & 30 \\ 33 & 33 \end{bmatrix} \\ \begin{bmatrix} 11 & 25 \\ 19 & 3 \end{bmatrix} & \begin{bmatrix} 22 & 39 \\ 14 & 30 \end{bmatrix} \end{pmatrix}$$

$$\begin{array}{r} 47 \\ 24 \\ \hline 71 \end{array}$$

1) $T(n) = 8T(n/2) + an^2$, $n > 2$ by Recurrence Method.

$$a = 8 \quad b = 2 \quad f(n) = an^2$$

$$T(n/2) = 8T(n/4) + a(n/2)^2$$

$$T(n/4) = 8T(n/8) + a(n/4)^2$$

$$T(n) = 8 \left[8T(n/4) + \frac{an^2}{4} \right] + an^2$$

$$= 8^2 T(n/4) + \frac{8^2 an^2}{4} + an^2$$

$$T(n) = 64T(n/4) + 3an^2$$

$$T(n/4) = 8T(n/8) + a(n/4)^2$$

$$T(n/8) = 8T(n/16) + a(n/8)^2$$

$$T(n) = 64 \left[8T(n/8) + \frac{an^2}{16} \right] + 3an^2$$

$$= 8^3 T(n/8) + \frac{64^4}{16} an^2 + 3an^2$$

$$T(n) = 8^3 T(n/8) + 4an^2 + 3an^2$$

$$T(n) = 8^3 T(n/8) + 3an^2 + 4an^2 = 8^3 T(n/8) + 7an^2$$

$$T(n) = 8^k T(n/2^k) + [an^2 + 3an^2 + 5an^2 + \dots] \text{ (or)}$$

$$T(n) = 8^k T(n/2^k) + an^2 [1 + 3 + 5 + 7 + \dots] \quad \begin{matrix} an^2 + 3an^2 + 5an^2 + 7an^2 \\ = an^2 [1 + 2 + 3 + \dots] \end{matrix}$$

$$n = 2^k$$

$$n = 2^k \Rightarrow k = \log_2 n \Rightarrow k = \log_2 n \quad = \frac{an^2 \cdot n(n+1)}{2}$$

$$= 8^{\log_2 n} T\left(\frac{2^k}{2^k}\right) + \underbrace{an^2 [1 + 3 + 5 + \dots]}_{\text{constant}} \quad = \frac{an^3 (n+1)}{2}$$

$$= n^3 + \text{constant}$$

$$T(n) = \Theta(n^3)$$

2) $T(n) = 7T(n/2) + an^2, n > 2$

$$a=7 \quad b=2 \quad f(n) = an^2$$

$$T(n/2) = 7T(n/4) + a(n/2)^2$$

$$T(n/2) = 7T(n/4) + \frac{an^2}{4}$$

$$T(n) = 7 \left[7T(n/4) + \frac{an^2}{4} \right] + an^2$$

$$T(n) = 7^2 T(n/4) + \frac{7an^2}{4} + an^2$$

$$T(n/4) = 7T(n/8) + a(n/4)^2$$

$$T(n/4) = 7T(n/8) + \frac{an^2}{16}$$

$$T(n) = 7^2 \left[7T\left(\frac{n}{8}\right) + \frac{an^2}{16} \right] + \frac{7}{4}an^2 + an^2$$

$$T(n) = 7^3 T\left(\frac{n}{8}\right) + \left(\frac{7}{4}\right)^2 an^2 + \left(\frac{7}{4}\right)an^2 + an^2$$

$$\vdots$$

$$T(n) = 7^k T\left(\frac{n}{2^k}\right) + an^2 \left[1 + \frac{7}{4} + \left(\frac{7}{4}\right)^2 + \dots \right]$$

$$n = 2^k \Rightarrow k = \log_2 n$$

$$= 7^{\log_2 n} T\left(\frac{2^k}{2^k}\right) + an^2 \left[\frac{\left(\frac{7}{4}\right)^k - 1}{\frac{7}{4} - 1} \right]$$

$$= 7^{\log_2 n} T(1) + an^2 \left[\frac{\left(\frac{7}{4}\right)^k - 1}{\frac{3}{4}} \right]$$

$$= n^{\log_2 7} T(1) + \frac{4}{3} an^2 \left[\frac{7^k - 4^k}{4^k} \right]$$

$$= n^{2.80} T(1) + \underbrace{\frac{4}{3} an^2 \left[\frac{7^k - 4^k}{4^k} \right]}_{\text{constant}}$$

$$T(n) = \Theta(n^{2.8})$$

→ QUICK SORT :-

i) In merge sort sub arrays or sub problems are sorted independently and later merged.
Whereas in quicksort the division of sub problems is made so that the sorted sub problems need not to be merged.

ii) Using divide & conquer strategy the quicksort uses partition of the elements and finding the solutions of element.

iii) Let $A[p \dots r]$ be an array which is divided into subproblems as $A[p] \dots a[q-1]$ $A[q]$ $A[q+1] \dots A[r]$ where $A[q]$ is pivot element

$\leq a[q]$

$\downarrow$
 pivot

 $\geq A[q]$

Conditions for quick sort :-

i)

$L \leftarrow R$
 $\uparrow$
 pivot

$p < r$?

if no → then swap pivot & right

$p < r$?

if yes → then move right one position towards left.

ii) $L \rightarrow R$
 $\uparrow$ pivot

If $p > \text{left}$?
 if No $\rightarrow$ then swap right & left.

If $p > \text{left}$?

If yes $\rightarrow$ then move left one position towards right.

*. Show how quick sort sorts the following sequence of keys.

1) 1, 1, 1, 1, 1, 1, 1 2) 5, 5, 8, 3, 4, 3, 2

1, 1, 1, 1, 1, 1, 1
 $\uparrow$ $\uparrow$
 P_L P_R
 $p < r$? no then swap

1, 1, 1, 1, 1, 1, 1
 $\uparrow$ $\uparrow$
 P_L P_R

1, 1, 1, 1, 1, 1, 1
 $\uparrow$ $\uparrow$
 P_L P_R

1, 1

ALGORITHM :-

Algorithm partition (a, m, p)

{

$v := a[m]; i := m; j := p;$

repeat

{

repeat

$i := i + 1;$

until $(a[i] \geq v);$

repeat

$j := j - 1;$

until $(a[j] \leq v);$

if $(i < j)$ then interchange $(a, i, j);$

}

until $(i \geq j);$

$a[m] := a[j];$

$a[j] := v;$

return j;

Algorithm Interchange(a, i, j)

{

$p := a[i];$

$a[i] := a[j];$

$a[j] := p;$

Algorithm Quicksort(p, q)

{

if $(p < q)$ then // if there are more than one element.

{

$j := \text{partition}(a, p, q+1);$ // j = pivot element

// divide p into sub problems

// j is the position of partition element.

Quicksort(p, j-1);

Quicksort(j+1, q);

}

}

When you go for analysis of quick sort we count only number of element comparison $C(n)$.

Best Case :- Quick sort. works best if each array is divided into two equal sub arrays of size $n/2$. This generates $\log(n)$ levels in the recursion tree. The recurrence relation for Best case is given by

$$T(n) = 2T(n/2) + cn \text{ when } n > 2.$$

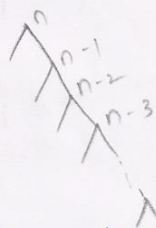
$$h(n) = \frac{cn}{n^{\log_2 2}} = c(1)^0 = c$$

$$\mu(n) = \theta(\log n)$$

$$T(n) = n^{\log_2 2} [T(1) + \mu(n)] = n' [\theta(\log n) + T(1)]$$

$$T(n) = \theta(n \log n)$$

Worst Case Analysis :- If the chosen pivot element is either smallest or largest in an array. In this case one part will be empty and other contains the remaining elements which generates $n-1$ levels in recursion tree.



If pivot is smallest element then the recursive relation is given as

$$T(n) = T(n-1) + cn \text{ where } n > 1$$

$$T(n) = T(n-1) + cn$$

$$T(n-1) = T(n-2) + c(n-1)$$

$$T(n-2) = T(n-3) + c(n-2)$$

$$T(n-3) = T(n-4) + c(n-3)$$

$$T(n) = T(n-1) + T(n-2) + T(n-3) + \dots + T(2) + T(1) + c(n-1) + c(n-2) + c(n-3) + \dots + c(2) + c(1)$$

$$T(n) = T(1) + c[1+2+3+\dots+n]$$

$$T(n) = T(1) + c \cdot \frac{n(n+1)}{2}$$

$$T(n) = c \cdot \frac{n(n+1)}{2} \quad (\because T(1) = \text{constant})$$

$$T(n) = n(n+1) \quad \because 1/2 \text{ is constant}$$

$$T(n) = n^2$$

$$T(n) = \Theta(n^2)$$

2) 5, 5, 8, 3, 4, 3, 2
 $\uparrow$
 P, L

$P < R$? NO

2, 5, 8, 3, 4, 3, 5
 $\uparrow$
 L

2, 5, 8, 3, 4, 3, 5
 $\uparrow$
 L

2, 5, 8, 3, 4, 3, 5
 $\uparrow$
 L

2, 5, 5, 3, 4, 3, 8
 $\uparrow$
 L

2, 5, 5, 3, 4, 3, 8
 $\uparrow$
 L

2, 5, 3, 3, 4, 5, 8
 $\uparrow$
 L

2, 5, 3, 3, 4, 5, 8
 $\uparrow$
 L

2, 5, 3, 3, 4, 5, 8
 $\uparrow$
 L

$P, L \downarrow$
 2, 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

$P, L \downarrow$
 2, 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

$P, L \rightarrow$ 2, 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

$P, L \rightarrow$ 2, 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

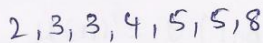
$P, L \rightarrow$ 2, 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

$\downarrow$
 5, 3, 3, 4 | 5, 8
 $\uparrow$
 R

$\downarrow$
 4, 3, 3, 5 | 5, 8
 $\uparrow$
 R

$\downarrow$
 4, 3, 3, 5 | 5, 8
 $\uparrow$
 R

$\downarrow$
 4, 3, 3, 5 | 5, 8
 $\uparrow$
 R


$$T_A(n) = (n+1) + \frac{1}{n} \sum_{1 \leq k \leq n} [T_A(k-1) + T_A(n-k)]$$

A - Average case

$$\sum_{1 \leq k \leq n} [T_A(k-1) + T_A(n-k)] \Rightarrow$$

$$k=n \Rightarrow T_A(n-1) + T'_A(0)$$

Multiply LHS & RHS by 'n'.

Put $n = n-1$ in equation (1)

$$\Rightarrow (n-1)T_A(n-1) = n(n-1) + 2[T_A(0) + T_A(1) + \dots + T_A(n-3) + T_A(n-2)] \quad \text{--- (2)}$$

$$\frac{T_A(n)}{n+1} = \frac{T_A(n-1)}{n} + \frac{T_A(n-2)}{n-1} + \frac{T_A(n-3)}{n-2} + \dots + \frac{T_A(1)}{2} +$$

$$\frac{2}{n+1} + \frac{2}{n} + \frac{2}{n-1} + \dots$$

$$\frac{T_A(n)}{n+1} = \frac{T(1)}{2} + 2 \left[\frac{1}{n+1} + \frac{1}{n} + \frac{1}{n-1} + \dots \right]$$

$$= 2 \sum_{2 \leq k \leq n+1} \frac{1}{k} \quad (\because T(n) \text{ is constant})$$

$$\frac{T_A(n)}{n+1} = 2 \int_3^{n+1} \frac{1}{k} dx$$

$$\frac{T_A(n)}{n+1} = 2 \left[\log k \right]_3^{n+1}$$

$$\frac{T_A(n)}{n+1} = 2 \left[\log(n+1) - \log 3 \right] = 2 \log n + 2 \log 1 - 2 \log 3$$

or

$$\frac{T_A(n)}{n+1} = 2 \left[\log(n+1) \right] \quad (\because \log 3 \text{ is constant})$$

$$T_A(n) = 2(n+1) \log(n+1)$$

$$T_A(n) = n \log(n+1) + \log(n+1)$$

$$T_A(n) = n \log(n+1) \quad \text{neglecting}$$

$$T_A(n) = n \log(n) \quad (\because 1 \text{ is constant and we can neglect it})$$

$$\therefore T_A(n) = O(n \log n)$$